Benchmark results

unknown AArch64 TOS 4.04 640x480 256c blit none

115 of 115 tests run, 1.0 s each. Values are operations per second (higher is better). Bars compare tests within each group.

Groups

VDI primitives	38 tests run
AES windows & events	21 tests run
Alignment (blitter)	20 tests run
Full-screen windows	17 tests run
GDOS fonts	14 tests run

VDI primitives

Test	ops/s	Note
Line, 2 points	52982	
Polyline, 32 points	2733.0	
Wide line (7px, round)	10526	
Dashed line	47260	
User line style	48664	
XOR line	53247	
Markers x16	42909	
Filled rect 32x32	51199	
Filled rect, full window	359.6	
Bar, hatched + border	32124	
Pattern fill rect	50711	
User pattern fill	50471	
XOR filled rect	81107	
Polygon, 8 points	23293	
Filled circle	4717.9	
Filled ellipse	4829.2	
Arc	60531	
Pie slice	4717.9	
Elliptical arc	59940	
Elliptical pie	6858.3	
Rounded box	93621	
Filled rounded box	16909	
Contour fill 64x40	3975.7	
Text, 15 chars	31506	
Text, 84 chars	7679.0	
Text with effects	14627	
Text rotated 90	24998	
Justified text	18339	
Text extent query	268100	
Blit scr->scr 64x64	13587	
Blit scroll window	278.6	
Blit mem->scr 64x64	33000	
Blit scr->mem 64x64	30116	
Mono blit (vrt_cpyfm)	13397	
Transform form 64x64	n/a	not supported here
Get pixel	355120	
Hide/show mouse	37066	
Attribute calls x3	127253	
Open/close workstation	24452	

AES windows & events

Test	ops/s	Note
Window create+delete	121662	
Window open+close	218.6	
Window move	795.2	
Window resize	1122.9	
Window top (2 windows)	442.5	
Set window title	3565.1	
Set info line	3583.0	
Set slider pos+size	6825.7	
Scroll window content	270.7	
wind_get work area	289128	
Rectangle list walk	129133	
wind_calc	240940	
wind_update pair	147455	
objc_draw dialog	5466.9	
objc_find	231848	
form_dial redraw 64x48	5466.9	
Menu bar redraw	6248.8	
graf_mkstate	290552	
graf_mouse off+on	31967	
Event poll (evnt_multi)	121662	
Message round trip	51948	

Alignment (blitter)

Test	ops/s	Note
Fill 64x64, aligned	15983	
Fill 64x64, x+3	16301	
Blit 64x64, aligned	12923	
Blit 64x64, src+3 dst+11	13114	
Mem->scr 64x64, aligned	32124	
Mem->scr 64x64, x+3	31506	
Mono blit, aligned	12923	
Mono blit, x+3	12677	
Text, x on 8 px	31355	
Text, x+3	30912	
Scroll up, aligned edges	306.7	
Scroll up, edges x+5	306.7	
Scroll left 16 px	314.2	
Scroll left 3 px	303.8	
Move window, aligned	663.2	work x mod 16 = 0
Move window, off-grid	657.0	work x mod 16 = 5
Open/close win, aligned	215.4	work x mod 16 = 0
Open/close win, off-grid	215.4	work x mod 16 = 5
Scroll window, aligned	988.4	work x mod 16 = 0
Scroll window, off-grid	983.6	work x mod 16 = 5

Full-screen windows

Test	ops/s	Note
Open/close full window	101.9	work 616x418
Full <-> normal size	198.0	
Top full / test window	240.5	work 616x418
Fill full window	236.6	work 616x418
Scroll up 1 line	176.0	work 616x418
Scroll left 16 px	175.0	work 616x418
Scroll left 8 px	176.0	work 616x418
Scroll left 3 px	176.8	work 616x418
Scroll text 8pt	176.8	system font
Scroll text 10pt	176.8	system font
Scroll text 12pt	177.6	got 10pt, system font
Scroll text 14pt	176.0	got 10pt, system font
Scroll text 18pt	174.1	system font
Scroll text 24pt	171.5	got 20pt, system font
Scroll text 36pt	170.7	got 20pt, system font
Scroll text 48pt	171.5	got 20pt, system font
Scroll, size per line	174.1	system font

GDOS fonts

Test	ops/s	Note
Text 6pt	48424	got 8pt, system font
Text 8pt	48424	system font
Text 10pt	22414	system font
Text 12pt	22527	got 10pt, system font
Text 14pt	22527	got 10pt, system font
Text 18pt	10477	system font
Text 24pt	5768.0	got 20pt, system font
Text 36pt	5878.4	got 20pt, system font
Text 48pt	5822.7	got 20pt, system font
Text 72pt	5850.4	got 20pt, system font
Text 10pt bold	21870	system font
vst_point, 8 sizes	273065	
Font switch + vst_point	n/a	not supported here
Arbitrary size + text	n/a	not supported here
Scroll, mixed sizes	262.1	system font
Text extent at 12pt	268100	
Enumerate font names	n/a	not supported here
Load+unload fonts (x3)	n/a	not supported here

Configuration

Machine

```
Description     : (not set - Options menu)
Machine type    : unknown
CPU             : AArch64 (native)
FPU             : none reported
Blitter         : not present
ST RAM          : 14336 KB total
TT/Fast RAM     : 925696 KB total
```

Operating system

```
TOS             : 4.04  (11/06/2026 m/d/y)
Country / video : USA, NTSC
GEMDOS          : 0.48
MiNT            : not running
Largest free    : ST 10240 KB
                  TT/Fast 914980 KB
```

AES / environment

```
AES version     : 3.40
Multitasking    : no (max apps 1)
Theme pens      : green 166, red 122, blue 149, cyan 155, highlight 106
Application id  : 0
appl_getinfo    : not available
Desktop area    : 0,19 640x461
AES char / box  : 8x16 / 19x19
Current dir     : C:\
PATH            : /dev/c/
```

Display

```
Resolution      : 640x480, Getrez 4 (TT Medium)
Colours         : 256 on screen, 16 planes
Palette         : 0 (direct colour)
Lookup table    : no (true colour)
Pixel size      : 278 x 278 microns
Screen memory   : 614400 bytes at $3c100000
VDI handles     : physical 1, ui 2 (bench ws opened only for runs)
VDI max vertices: 1024
Text rotation   : 90 degree steps
Writing modes   : 4
```

GDOS / fonts

```
GDOS            : none (system font only)
Fonts           : 1 (0 loaded by GDOS)
  > id     1  6x6 system font
```

Cookie jar

```
NTOS _MCH _IDT _AKP EdDI
```